

Algorithms 2

Graph Colouring

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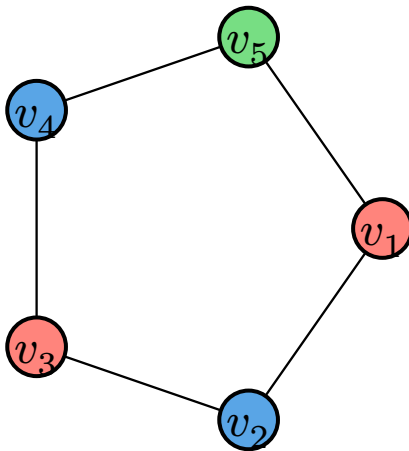
1. A Greedy Colouring Algorithm

- A Greedy Colouring Algorithm:
 - Input:
 - A graph G
 - An ordering $v_1, v_2, \dots, v_n$ of $V(G)$
 - Output:
 - A legal vertex-colouring of G
1. Scan the vertices according to the ordering.
 2. When considering v_i , assign it the least available colour.
 3. This always produces a legal colouring.

Question 1.1.1 How useful is this algorithm for bounding $\chi(G)$?

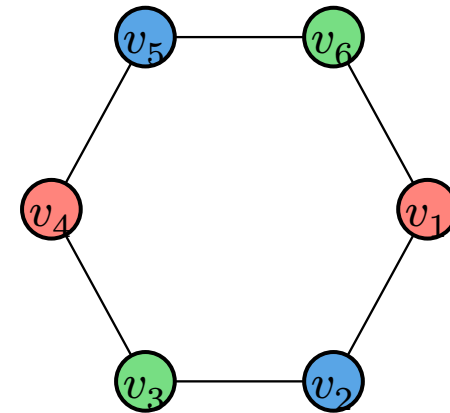
Question 1.2.1 How useful is this algorithm for bounding $\chi(G)$?

C_5 (greedy uses 3 colours)



v_1, v_2, v_3, v_4, v_5
1, 2, 1, 2, 3

C_6 (greedy can also use 3 colours)



$v_1, v_4, v_2, v_3, v_5, v_6$
1, 1, 2, 3, 2, 3

- For every graph G , define

$$\Delta(G) := \max\{\deg_{G(v)} : v \in V(G)\}$$

- The greedy algorithm implies $\chi(G) \leq \Delta(G) + 1$
- When the bound is tight?
 - There are graphs with

$$\chi(G) = \Delta(G) + 1$$

- Examples:
 - Odd cycles: $\Delta = 2$ and $\chi = 3$
 - Complete graphs: $\Delta(K_r) = r - 1$ and $\chi(K_r) = r$

2. Brooks' Theorem

Theorem 2.1.1 (Brooks' Theorem) Let G be a connected graph. Then

$$\chi(G) \leq \Delta(G)$$

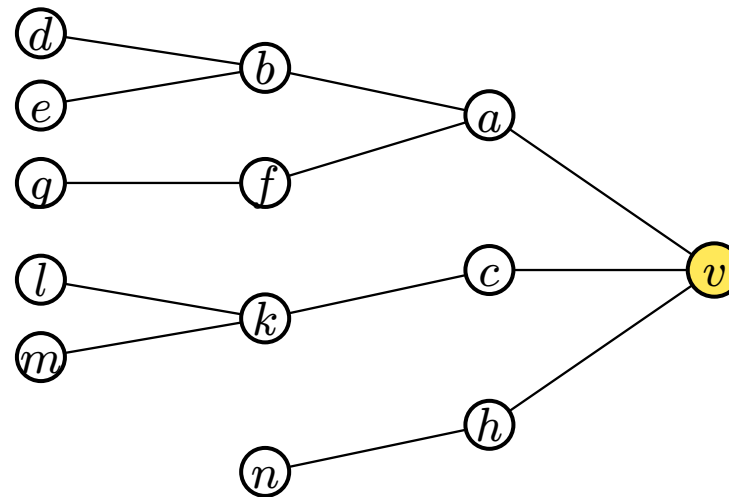
unless G is complete or an odd cycle.

Observation 2.1.2 Brooks' upper bound can be far from optimal in general.

proof:

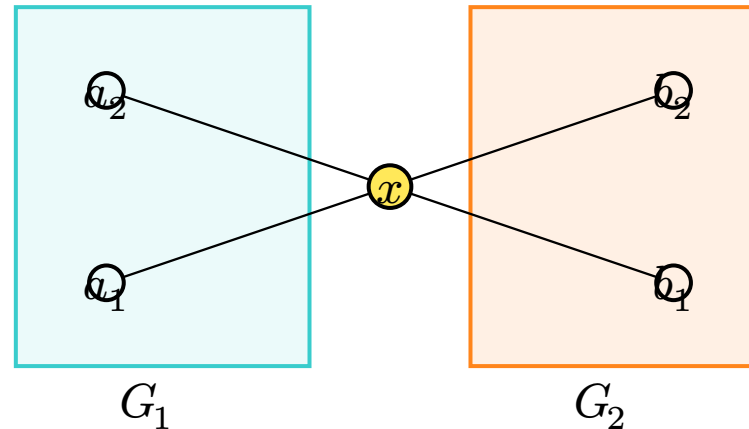
- Assume $\Delta(G) \geq 3$.
- If $\Delta(G) \leq 2$, then connected G is a path, cycle, or odd cycle, so the theorem is immediate.
- Distinguish two cases:
 - There exists $v \in V(G)$ with $\deg(v) < \Delta(G)$.
 - Otherwise G is $\Delta(G)$ -regular.

- Choose $v \in V(G)$ with $\deg(v) < \Delta(G)$.
- Let T be a spanning tree of G rooted at v .
- Order vertices so descendants appear before ancestors and v is last.
- In this ordering, each vertex except possibly v has at most $\Delta(G) - 1$ earlier neighbours.
- Greedy colouring then yields a $\Delta(G)$ -colouring.



Example descendants-first order: $d, e, b, q, f, a, l, m, k, c, n, h, v$.

- Assume G is $\Delta(G)$ -regular.
- If $\kappa(G) = 1$, then G has a cut-vertex.
- Colour components around the cut-vertex separately and permute colour names to combine them.



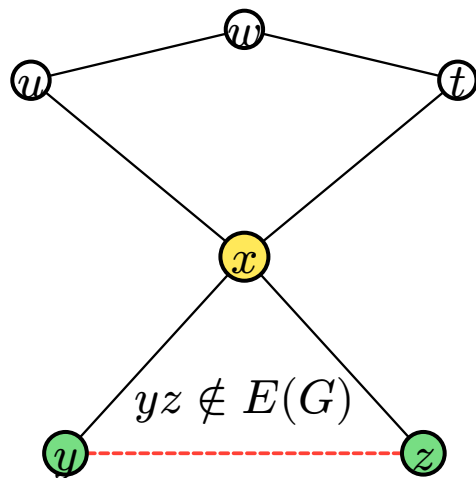
- Therefore we may reduce to the 2-connected case:

$$\kappa(G) \geq 2$$

- It is enough to find vertices $x, y, z \in V(G)$ such that:
 - $xy, xz \in E(G)$
 - $yz \notin E(G)$
 - $G - \{y, z\}$ is connected

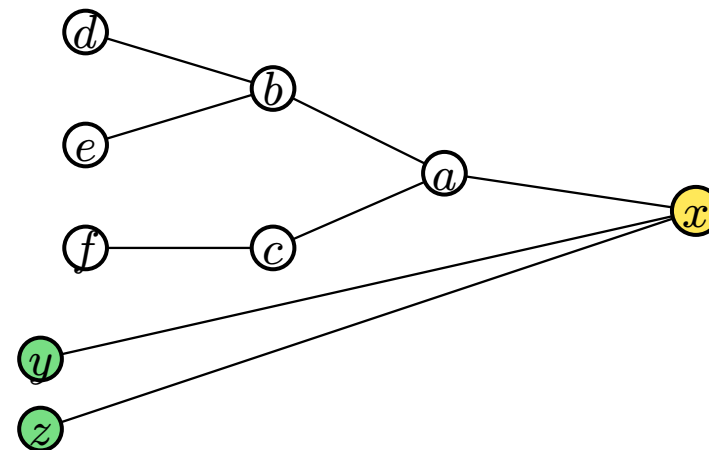
- If such vertices exist:
 - Build a spanning tree of $G - \{y, z\}$ rooted at x .
 - Use an order with x last and y, z first.
 - Since y and z are nonadjacent, they may receive the same colour.
 - Greedy colouring then gives a legal $\Delta(G)$ -colouring.

Cherry condition



$xy, xz \in E(G)$ and $G - \{y, z\}$ stays connected.

Rooted tree for greedy order

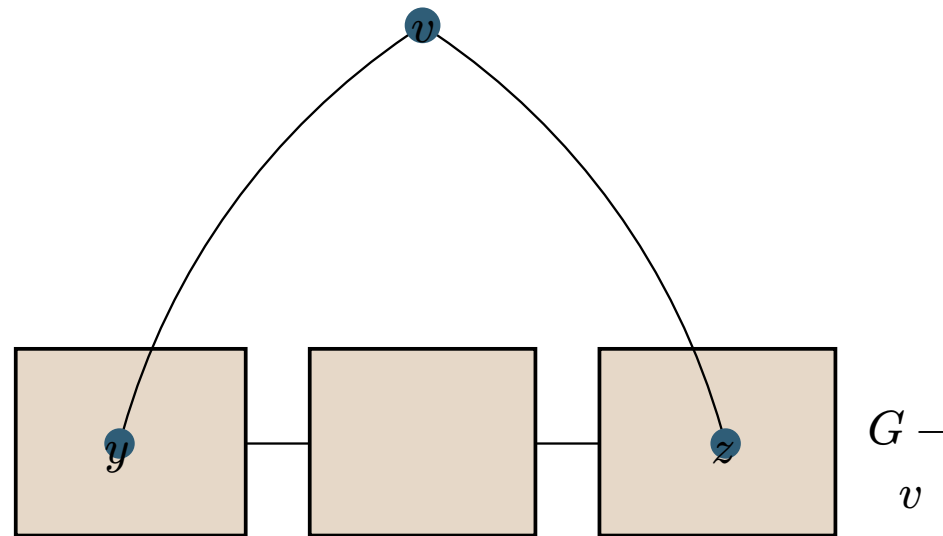


Order example: $y, z, d, e, b, f, c, a, x$ (with x last).

Lemma 2.5.1 (Structural Lemma) Let $3 \leq k \in \mathbb{N}$ and let G be a k -regular, 2-connected, non-complete graph. Then there exist $x, y, z \in V(G)$ such that:

1. $xy, xz \in E(G)$
2. $yz \notin E(G)$
3. $G - \{y, z\}$ is connected.

- First subcase: there exists v such that $G - v$ is connected but not 2-connected.
 - Analyze the block tree of $G - v$.
 - Pick y, z in different leaf blocks, and set $x := v$.
 - Then $G - \{y, z\}$ stays connected.



Block decomposition of $G - v$: blocks are separated; red cut-vertices in gaps; y, z in leaf blocks, edges from v to leaves.

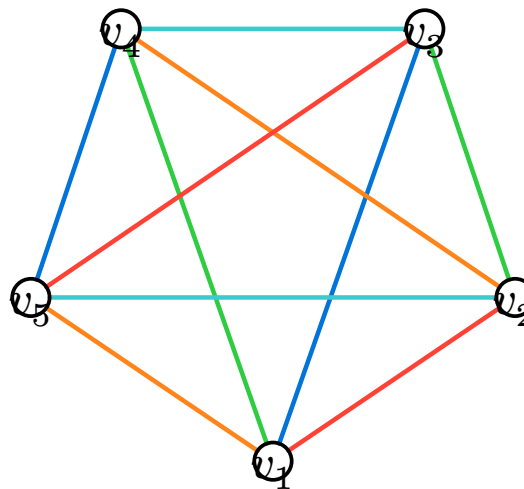
- Second subcase: $\kappa(G - v) \geq 2$ for every $v \in V(G)$.
 - Since G is not complete, pick nonadjacent y, z among neighbours of some x .
 - $\kappa(G - y) \geq 2$
 - Connectivity conditions force $G - \{y, z\}$ to remain connected.

3. Edge Colouring

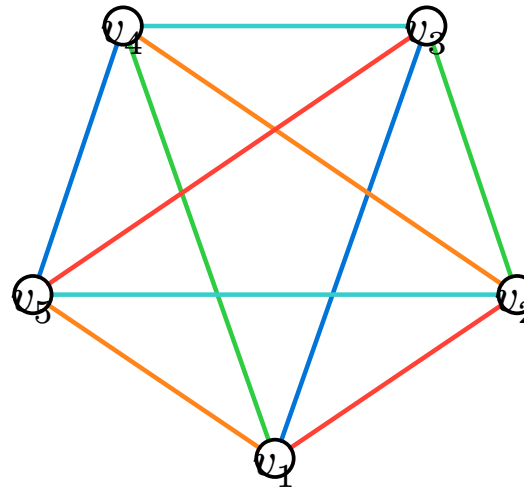
Definition 3.1.1 An edge-colouring of a graph G with k colours is a function

$$\varphi : E(G) \rightarrow [k]$$

Definition 3.1.2 An edge-colouring is called proper if incident edges receive distinct colours.



Definition 3.1.3 The chromatic index of G , denoted by $\chi'(G)$, is the least k such that G has a proper edge-colouring with k colours.



- here $\chi'(G) = 5$.

Definition 3.2.1 The line graph of G , denoted $L(G)$, is the graph with:

- $V(L(G)) = E(G)$
- two vertices adjacent iff the corresponding edges of G are incident.

Observation 3.2.2

- $\chi(L(G)) = \chi'(G)$
- $\Delta(G) \leq \chi'(G)$
- $\Delta(L(G)) \leq 2(\Delta(G) - 1)$

- Brooks tells us:

$$\chi'(G) = \chi(L(G)) \leq 2(\Delta(G) - 1)$$

4. Konig's Theorem

Theorem 4.1.1 (König's Theorem) If G is bipartite, then

$$\chi'(G) = \Delta(G)$$

- Remember that $\chi'(G) \geq \Delta(G)$ always holds, so we only need to show $\chi'(G) \leq \Delta(G)$ for bipartite graphs.

proof:

- Degree-regular bipartite graphs admit a perfect matching (See in the practice session).
- Removing edges of a perfect matching from a k -regular bipartite graph gives a $(k - 1)$ -regular bipartite graph.
- By induction on k , every k -regular bipartite graph is k -edge-colourable.
- Any bipartite graph can be embedded into a $\Delta(G)$ -regular bipartite supergraph.
- Restrict the colouring from the supergraph back to G .

5. Vizing's Theorem

Theorem 5.1.1 (Vizing's Theorem) For every simple graph G ,

$$\chi'(G) \leq \Delta(G) + 1$$

- Combined with the trivial lower bound:

$$\Delta(G) \leq \chi'(G) \leq \Delta(G) + 1$$

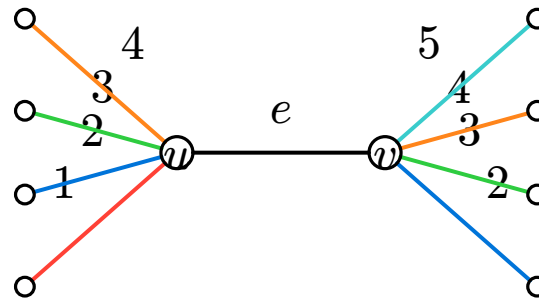
- So every graph is either:
 - Class 1 ($\chi'(G) = \Delta(G)$)
 - Class 2 ($\chi'(G) = \Delta(G) + 1$).
 - It is NP-hard to determine the class of a given graph.

Theorem 5.2.1 (Vizing's Theorem) For every simple graph G ,

$$\chi'(G) \leq \Delta(G) + 1$$

proof:

- Proof by induction on $e(G)$.
- Pick $e = uv \in E(G)$ and set $G' := G - e$.
- By induction hypothesis, G' has a proper edge-colouring with at most $\Delta(G) + 1$ colours.
- Try to colour e when re-inserting it.



- Each endpoint misses at least one colour among $[\Delta(G) + 1]$.
- If there is a common missing colour at both u and v , assign it to e .
- Otherwise, what can we do?

- Each endpoint misses at least one colour among $[\Delta(G) + 1]$.
- If there is a common missing colour at both u and v , assign it to e .
- Otherwise, what can we do?
- That's up for the TA to show you.